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Hiranaka

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(54) **CRYSTALLIZATION METHOD OF GYPSUM**

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C04B 11/06; C04B 11/002; C04B 11/005;
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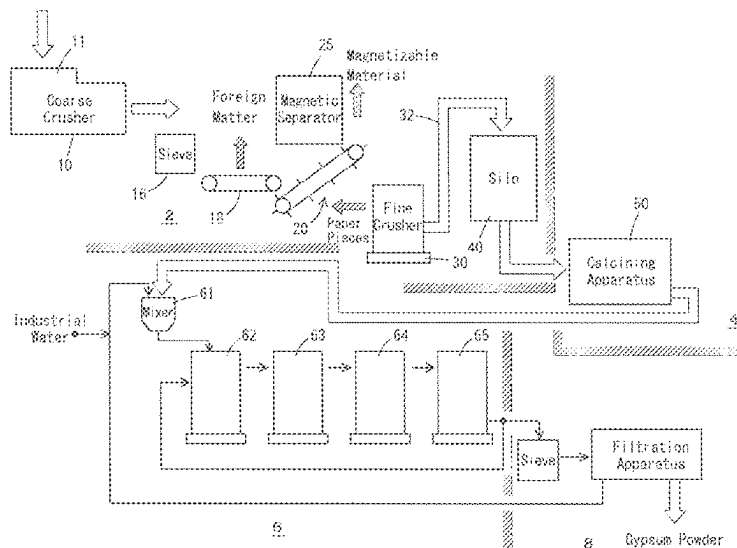
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(57) **ABSTRACT**

Wasted gypsum boards are crushed and calcined to gypsum granular solid, and the gypsum granular solid is mixed with water to form gypsum slurry. Gypsum particles are deposited from the gypsum slurry in a crystallization tank. Heated steam is blown into the gypsum slurry at a height from surface of the gypsum slurry and down to upper 1/3 of the gypsum slurry to heat the gypsum slurry and to eliminate foam on the gypsum slurry.

5 Claims, 2 Drawing Sheets



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See application file for complete search history.

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FIG. 1

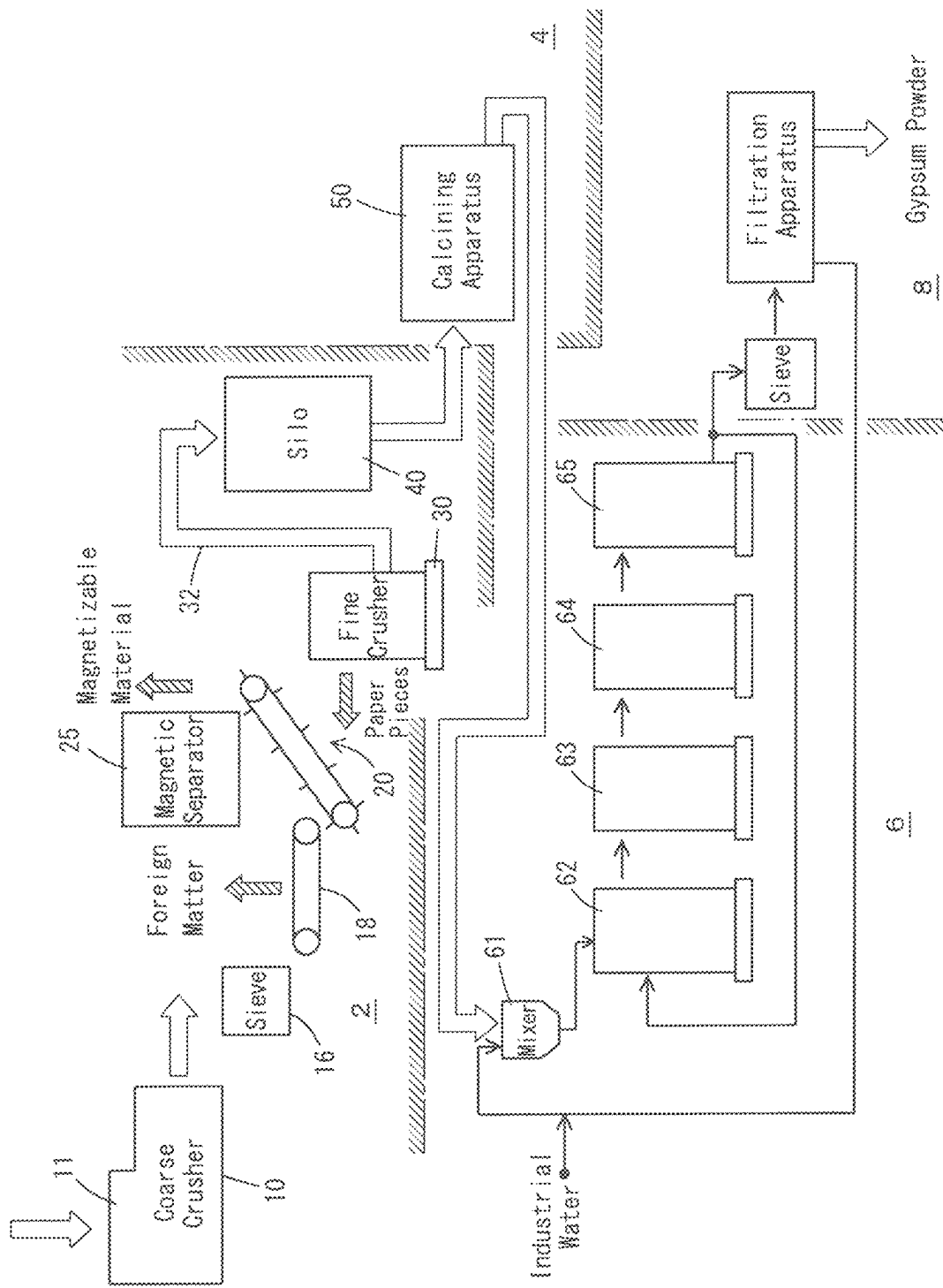
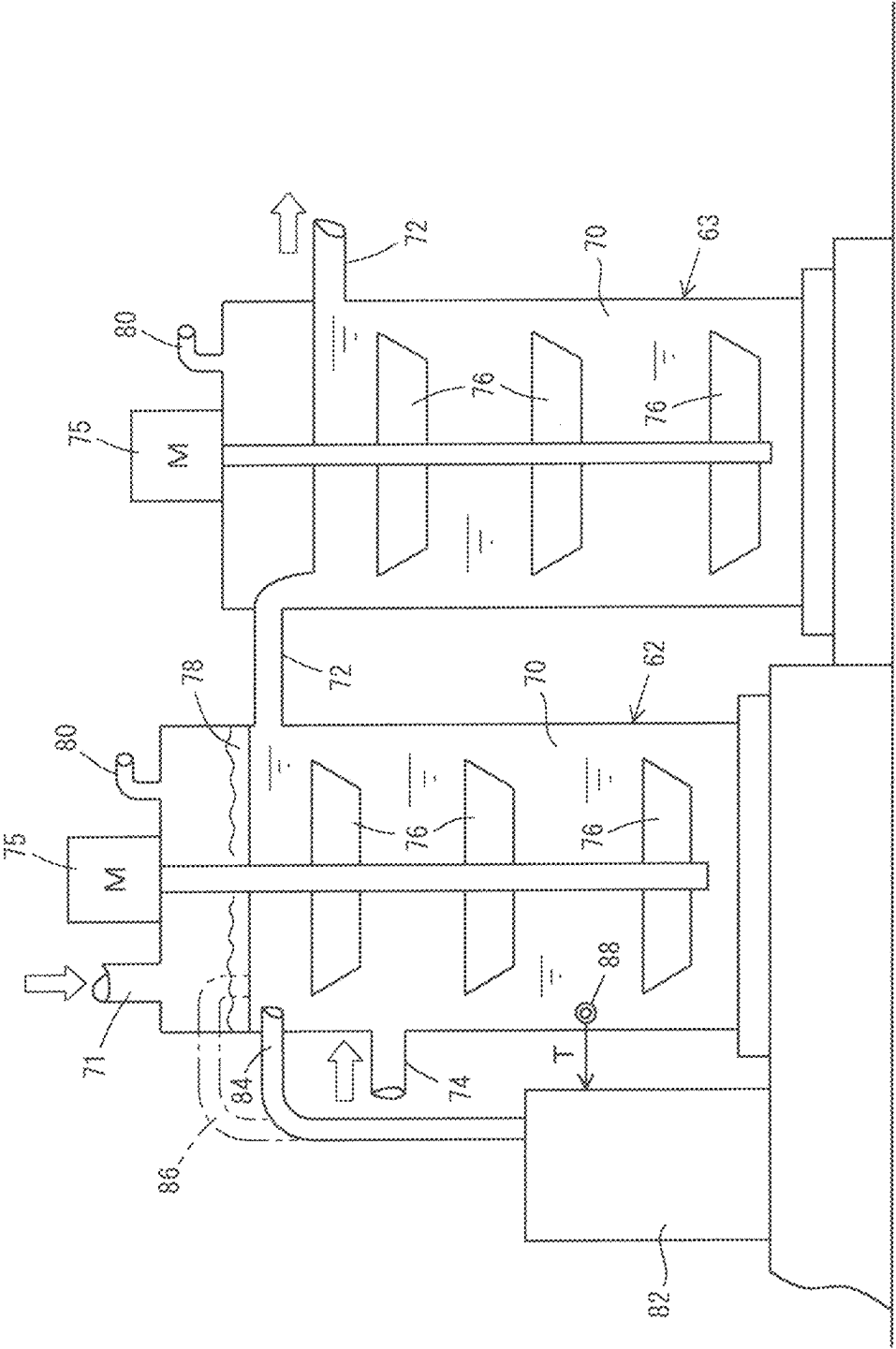


FIG. 2



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CRYSTALLIZATION METHOD OF GYPSUM**TECHNICAL FIELD**

The present invention relates to a crystallization method 5
of gypsum.

BACKGROUND ART

The present inventor and others have proposed to reclaim 10
gypsum from wasted gypsum boards (e.g., Patent Document
1: WO2012/176688). The wasted gypsum boards are
crushed by a crusher and calcined to hemihydrate and/or
anhydrous Type III gypsum granular solid. The resultant
gypsum is mixed with water, etc., and particles of dihydrate 15
gypsum and so on are deposited in a crystallization tank, and
then, gypsum is reclaimed by solid-liquid separation.

The type of gypsum deposited in the crystallization tank
varies depending on the slurry temperature. For example,
dihydrate gypsum is deposited at temperatures up to 90 20
degree Celsius, and hemihydrate gypsum and/or anhydrous
type III gypsum are deposited at higher temperatures. In the
range from room temperature to 90 degree Celsius, the
higher the temperature, the deposited dihydrate gypsum
becomes larger in grain size and has a higher industrial
value. For this reason, the gypsum slurry in the crystalliza-
tion tank is heated, but the heating is costly.

Wasted gypsum boards contain surfactants and so on.
They cause foaming and make gypsum slurry overflow from
the crystallization tank when mixing gypsum with the slurry.
Further, the foam covering the gypsum granular solid delays
the dissolution of the gypsum granular solid into the slurry.

PRIOR DOCUMENT LIST

Patent Document

Patent document 1: WO2012/176688

SUMMARY OF THE INVENTION**Problem to be Solved**

The object of the present invention is to reduce the cost of
heating gypsum slurry and to eliminate the foam in the 45
crystallization tank when crystallizing the gypsum slurry
derived from wasted gypsum boards.

Means for Solving the Problem

Wasted gypsum boards are crushed and calcined to gyp-
sum granular solid, the gypsum granular solid is mixed with
water to form gypsum slurry, and gypsum particles are
deposited from the gypsum slurry in a crystallization tank.

According to the present invention, heated steam is blown 55
into the gypsum slurry at a height from the surface of the
gypsum slurry and down to the upper $\frac{1}{5}$ of the gypsum
slurry for heating the gypsum slurry, and for eliminating
foam on the gypsum slurry.

Blowing the heated steam directly in the gypsum slurry 60
has a higher thermal efficiency than flowing a heated thermal
medium through a heater jacket enclosing the crystallization
tank for indirect heating. More, the heated steam blown in
the upper portion of the gypsum slurry heats, expands, and
breaks the foam on the gypsum slurry. Therefore, the heating
and defoaming of the gypsum slurry are carried out simul-
taneously. 65

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For maintaining the gypsum slurry at a predetermined
temperature, in addition to the heating by the steam, a heater
jacket may be provided to cover the crystallization tank and
to circulate a thermal medium within the heater jacket.
However, it is preferable to maintain the gypsum slurry at
the predetermined temperature without another thermal
source provided in the crystallization tank for reducing the
heating cost.

Preferably, the crystallization tank comprises plural crys-
tallization tanks in series, and the gypsum slurry resultant
by mixing the gypsum granular solid with water is supplied
into the most upstream crystallization tank. The gypsum slurry
is moved from an upstream tank to a downstream tank in
series, and the heated steam is only blown into the most
upstream crystallization tank. The heated steam blown into
the most upstream crystallization tank maintains the slurry
temperature in the downstream crystallization tank near the
predetermined temperature. Since the foam is generated in
the most upstream crystallization tank, the elimination of the
foam in the most upstream tank is enough.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1: A summary diagram indicating the reclaiming
method of gypsum from wasted gypsum boards.

FIG. 2: A schematic vertical cross-sectional view of the
crystallization tanks according to the embodiment.

**FEATURES FOR CARRYING OUT THE
INVENTION**

One Embodiment for carrying out the present invention
will be described. The scope of the present invention shall be
determined based on the claims and in accordance with the
understanding of those skilled in the art, taking into con-
sideration the description of the specification and well-
known art in the field. The scope of the present invention is
not limited by the embodiment.

Embodiment

FIGS. 1 and 2 show the embodiment. FIG. 1 shows the
method from crushing wasted gypsum boards to reclaiming
of new gypsum. In the first pretreatment process 2, wasted
gypsum boards, not shown in the figure, are coarsely
crushed by a coarse crusher 10 through an inlet port 11. The
coarsely crushed pieces are separated by a sieve 16 to
gypsum powder as the under-sieve component. The remain-
ing over-sieve component is conveyed on a sorting conveyor
18, and foreign matters such as metal, wood chips, mortar,
etc. are removed manually or automatically. After removing
the foreign matters, the crushed pieces and so on are
conveyed by a metering conveyor 20, and magnetizable
matters such as metals are separated by a magnetic separator
25. Then, gypsum pieces are crushed into fine gypsum
granular solid by a fine crusher 30, for example, and paper
pieces are further removed by a built-in sieve in the fine
crusher 30. Then, the gypsum granular solid is further
conveyed by an air current through a magnetic separation
pipe 32, and small metal objects, such as screws and nails,
are separated by magnets installed in the magnetic separa-
tion pipe 32. The gypsum granular solid passed through the
magnetic separation pipe 32 is stocked in a silo 40.

In the next calcining process 4, gypsum granular solid is
calcined by a calcining apparatus 50 to convert them to
hemihydrate and/or anhydrous type III gypsum. The calc-
ined gypsum granular solid is crystallized in the third

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crystallization process **6**. First, the calcined gypsum granular solid is mixed with an aqueous medium, such as gypsum slurry, in a mixer **61**. Gypsum is then crystallized from the slurry in crystallization tanks **62-65**, for example, in a series of 4 stages, and dihydrate gypsum particles are deposited. As a remark, when the temperature of crystallization tanks **62-65** is increased above 90 degree Celsius, hemihydrate gypsum or the like is deposited. The crystallization tank **62** is the most upstream tank, and the numbers **62-65** indicate the order from the upstream to the downstream tanks. The gypsum slurry is circulated among the crystallization tanks **62-65**. In the embodiment, the gypsum slurry is returned from the crystallization tank **65** to the crystallization tank **62**, but the circulation of the gypsum slurry may be omitted.

The gypsum granular solid may be fed into the aqueous medium flowing through a chute or a similar device without using the mixer **61**. A single-stage large crystallization tank may be used instead of the multi-stage crystallization tanks **62-65**.

In the filtration process **8**, the gypsum slurry is extracted from the crystallization tank **65**, etc., and sieved to remove paper fragments. The gypsum slurry is then solid-liquid separated in a filtration apparatus to extract gypsum particles such as dihydrate gypsum particles. The remaining liquid component is circulated, for example, to the mixer **61**. Industrial water is added to replenish the water lost as the crystallization water and adhering water of the gypsum particles.

FIG. **2** shows the crystallization tanks **62** and **63**, and the third and fourth crystallization tanks **64** and **65** are configured in the same manner as the second stage crystallization tank **63**. Gypsum slurry **70** is stored in the crystallization tanks **62** and **63**, and the gypsum slurry is supplied from a slurry supply pipe **71** connected to the mixer into the tank **62**. A circulation pipe **72** connects the crystallization tanks **62**, **63**, and so on, and moves the gypsum slurry to the downstream tanks **63-65** by natural flow, for example. The gypsum slurry is circulated to the most upstream tank **62** by a circulation pipe **74** from the most downstream tank **65**. Each of the crystallization tanks **62-65** is provided with vanes **76** and a driving motor **75** to stir the gypsum slurry **70**. When the slurry is made from hemihydrate and/or anhydrous Type III gypsum derived from the wasted gypsum boards, foam **78** is generated, especially in the most upstream crystallization tank **62** due to surfactants and so on in the wasted gypsum boards.

At or just under the slurry level of the most upstream crystallization tank **62**, generally, at a height down to the upper $\frac{1}{5}$ from the surface of the slurry **70**, an outlet of a steam introduction pipe **84** is placed to blow heated steam into the gypsum slurry **70**. Steam may be blown into the gypsum slurry from just over the slurry level as in the steam introduction pipe **86** shown by the chain line. Indicated by **82** is a steam generator such as a boiler. A temperature sensor **88** measures the temperature **T** of the gypsum slurry **70** to control the steam generator **82**.

The heated steam blown in from the steam inlet pipes **84** or **86** is, for example, at 130 degree Celsius and with 1 atmospheric pressure, and so on. Steam with 0.2 to 2 atmospheric pressure and at 100 to 150 degree Celsius is preferable for easy operation. The pressure here indicates the difference from the outside atmospheric pressure, and 0.2 to 2 atmospheric pressures indicate 1.2 to 3 atmospheric pressures of pressurized steam in the absolute pressure. The heated steam is, in most cases, superheated steam having a temperature over 100 degree Celsius. When the gypsum slurry in the most upstream tank **62** is made at just under 90

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degree Celsius, all the crystallization tanks **62-65** are maintained at the required temperature, except in cold climates. It is less expensive to blow the heated steam directly into the crystallization tank **62** than to heat indirectly the crystallization tank **62** by a heater jacket or other means.

The heated steam blown in the slurry heats the slurry near the slurry surface, expands, and breaks the foam **78** as a result. Therefore, the foam **78** is eliminated at the same time as the gypsum slurry **70** is heated. The blowing of the heated steam dilutes the gypsum slurry, but it is usually less than the water exhausted as the crystallization water and the adhering water of the deposited dihydrate gypsum. Therefore, except in cold climates, a heating jacket or a similar device is not needed other than the heated steam.

LIST OF SYMBOLS

- 2** pretreatment process
- 4** calcining process
- 6** crystallization process
- 8** filtration process
- 10** coarse crusher
- 11** inlet port
- 16** sieve
- 18** sorting conveyor
- 20** metering conveyor
- 25** magnetic separator
- 30** fine crusher
- 32** magnetic separation pipe
- 40** silo
- 50** calcining apparatus
- 61** mixer
- 62-65** crystallization tank
- 70** gypsum slurry
- 71** slurry supply pipe
- 72, 74** circulation pipe
- 75** motor
- 76** vane
- 78** foam
- 80** ventilation hole
- 82** steam generator
- 84, 86** steam introduction pipe
- 88** temperature sensor

The invention claimed is:

1. A method of reclaiming gypsum particles from wasted gypsum boards, the method comprising:
 - crushing and calcining wasted gypsum boards to gypsum granular solid,
 - mixing the gypsum granular solid with water to form gypsum slurry in a mixing tank,
 - generating gypsum particles in the gypsum slurry in at least a crystallization tank wherein surfactants from the wasted gypsum boards cause foam on the gypsum slurry, and
 - blowing heated steam into the gypsum slurry in the crystallization tank at a height between a top surface of the gypsum slurry and $\frac{1}{5}$ below the top surface of the gypsum slurry so that the gypsum slurry is heated only by the heated steam, and the foam on the gypsum slurry is eliminated by the heated steam.
2. The method of reclaiming gypsum particles according to claim 1, wherein the gypsum slurry is maintained at a predetermined temperature without a thermal source, other than the heated steam, provided in said at least a crystallization tank.
3. The method of reclaiming gypsum particles according to claim 1,

wherein said at least a crystallization tank comprises plural crystallization tanks in series, the gypsum slurry which is resultant by mixing the gypsum granular solid with water is supplied into a most upstream crystallization tank, the gypsum slurry is moved from an upstream tank to a downstream tank, and the heated steam is only blown into the most upstream crystallization tank. 5

4. The method of reclaiming gypsum particles according to claim 1, wherein said heated steam at a point where the heated steam is blown into the gypsum slurry is at 100 to 150 degree Celsius and has 1.2 to 3 atmospheric pressure in absolute pressure. 10

5. The method of reclaiming gypsum particles according to claim 4, wherein said heated steam heats the gypsum slurry to a temperature from above room temperature to not higher than 90 degree Celsius. 15

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